

FULL-LENGTH ORIGINAL RESEARCH

Epilepsia®

Determinants of IQ outcome after focal epilepsy surgery in childhood: A longitudinal case-control neuroimaging study

Caroline Skirrow^{1,2,3}  | J. Helen Cross^{1,4} | Rosie Owens^{1,2} | Louise Weiss-Croft^{1,2,5} | Patricia Martin-Sanfilippo^{1,2} | Tina Banks^{1,6} | Emily Shah¹ | William Harkness^{1,7} | Faraneh Vargha-Khadem^{1,2} | Torsten Baldeweg^{1,2} 

¹Developmental Neurosciences Programme, Great Ormond Street Institute of Child Health, University College London, London, UK

²Department of Neuropsychology, Great Ormond Street Hospital for Children NHS Foundation Trust, London, UK

³Cambridge Cognition, Cambridge, UK

⁴Department of Neurology, Great Ormond Street Hospital for Children NHS Foundation Trust, London, UK

⁵Science Gallery London, King's College London, London, UK

⁶Department of Radiology, Great Ormond Street Hospital for Children NHS Foundation Trust, London, UK

⁷Department of Neurosurgery, Great Ormond Street Hospital for Children NHS Foundation Trust, London, UK

Correspondence

Torsten Baldeweg and Caroline Skirrow,
Developmental Neurosciences Programme,
Great Ormond Street Institute of Child
Health, University College London,
London, UK.

Emails: t.baldeweg@ucl.ac.uk;
c.skirrow@ucl.ac.uk

Funding information

This research was funded by Epilepsy Research UK and the Great Ormond Street Hospital Children's Charity and supported by the National Institute for Health Research Biomedical Centre at Great Ormond Street Hospital for Children National Health Service Foundation Trust and University College London. The funding bodies provided financial support for this study but had no input into study design and analysis. The corresponding authors had full access to all the data and had final responsibility for the decision to submit for publication

Summary

Objective: Intelligence quotient (IQ) outcomes after pediatric epilepsy surgery show significant individual variation. Clinical factors such as seizure cessation or antiepileptic medication discontinuation have been implicated, but do not fully account for the heterogeneity seen. Less is known about the impact of neurobiological factors, such as brain development and resection location. This study examines clinical and neuroimaging factors associated with cognitive outcome after epilepsy surgery in childhood.

Methods: Fifty-two children (28 boys, 24 girls) were evaluated for epilepsy surgery and reassessed on average 7.7 years later. In the intervening time, 13 were treated pharmacologically and 39 underwent focal surgery (17 temporal, 16 extratemporal, six multilobar; mean age at surgery = 14.0 years). Pre- and postsurgical assessments included IQ tests and T1-weighted brain images. Predictors of IQ change were investigated, including voxel-based analyses of resection location, and gray and white matter volume change.

Results: Overall modest IQ improvement was seen in children treated surgically, but not in those treated pharmacologically only. Applying a ≥ 10 -point change threshold, 39% of the surgically treated children improved, whereas 10% declined. Clinical factors associated with IQ increases were lower preoperative IQ and longer follow-up duration, whereas seizure and antiepileptic medication cessation were not predictive. Among neuroimaging factors, we observed that left anterior temporal resections impacted negatively on verbal reasoning, linked to full-scale IQ decline. In contrast, gray matter volume change in ipsi- and contralesional hemispheres was positively

correlated with IQ change. Voxel-based morphometry identified the gray matter volume change in the contralesional dorsolateral frontal cortex as most strongly associated with IQ improvement.

Significance: We show that a variety of factors are likely to contribute to patterns of postsurgical change in IQ. Neuroimaging results indicate that left anterior temporal resections constrain development of verbal cognition, whereas simultaneously cortical growth after surgical treatment can support improvements in IQ.

KEYWORDS

intelligence quotient, longitudinal, neuroimaging, neurosurgery, outcomes, seizures

1 | INTRODUCTION

Resective surgery is a treatment for severe, medication-resistant epilepsy that aims to cure seizures by localizing and safely removing a critical mass of the epileptogenic network in the brain.¹ Around 60% of children are free from seizures at 5 or more years after temporal or extratemporal surgery.² Research indicates that in children epilepsy surgery overall has either little impact,^{3–7} or possibly a positive effect^{8–12} on cognitive function, as measured by intelligence quotient (IQ). However, embedded in these group effects is significant interindividual variation. Recent systematic reviews of studies reporting IQ outcomes after temporal and extratemporal lobe surgery found that 21%–30% of patients improved, 58%–67% remained stable, and 10%–12% deteriorated.^{13,14} This begs the question: why do some children show improvement but others decline?

Factors that are likely to contribute include postsurgical seizure freedom,^{3,8} cessation of antiepileptic drug (AED) use,^{4,9,15} preoperative IQ,^{8,9,16} and duration of follow-up.^{8,9} Less attention has been paid to how brain structural changes and postsurgical brain development are linked to cognitive function. Epilepsy and refractory seizures are associated with reductions in brain volume,^{17,18} as well as abnormal structural brain development.¹⁹ These findings may indicate seizure-induced brain growth suppression or bilaterally acquired cortical damage caused by seizures.¹⁸ How these effects are modified by surgery requires investigation.

One study that has evaluated this association found improved full-scale IQ (FSIQ) after childhood temporal lobe surgery with concurrent change in global brain gray matter (GM) volumes, with greater improvements associated with more positive volumetric changes.⁹ In this same sample, better semantic memory outcome was associated with greater postsurgical temporal pole integrity.²⁰ Surgery may therefore simultaneously constrain and enhance cognitive development.

Key Points

- Children with temporal and extratemporal surgery for epilepsy overall show modestly improved IQ at 2–16 years postsurgery
- However, there is significant individual variation in IQ outcomes after pediatric epilepsy surgery
- Postsurgical gray matter growth is associated with IQ improvements
- Left anterior resections may constrain the development of verbal cognition

The current study follows a longitudinal case-control design, examining IQ change after childhood temporal and extratemporal surgery, compared to an epilepsy control group with similar clinical characteristics. We aim to identify factors associated with change in IQ. We expect to find postsurgical IQ increase in the surgically treated sample, and will examine the role of the following predictors of postsurgical cognitive change: (1) clinical contributors (such as epilepsy and medication factors), (2) side and location of surgery, and (3) brain growth. Further exploratory analyses attempted to localize brain volume changes associated with IQ change, and identify the most robust predictors of postsurgical IQ change.

2 | MATERIALS AND METHODS

2.1 | Participants

Children with epilepsy who underwent presurgical cognitive evaluation between 1999 and 2013 in the neuropsychology department at Great Ormond Street Hospital for Children (GOSH) were identified from hospital notes. Children with focal epilepsy and surgery, and children with a history of medication-resistant epilepsy who had

not undergone surgical treatment were recruited into the study. Participants were selected on the basis that they were able to take part in magnetic resonance imaging (MRI) and cognitive assessments. Inclusion criteria were medication-resistant epilepsy (inadequate seizure control after treatment with ≥ 2 AEDs), preoperative IQ assessments available at age ≥ 5 years, and either focal resective surgery at GOSH or the National Hospital of Neurology and Neurosurgery (NHNN), or continued treatment with antiepileptic drugs. Exclusion criteria were nonfluent in English, hemispherectomy/hemispherotomy, multiple lesionectomies or disconnections, neural implant precluding MRI (shunt or vagus nerve stimulation device), and acquired head injury after surgery. Hospital records identified 61% as having undergone surgery, 30.5% as being treated nonsurgically, and 8.5% as without clear documentation regarding surgical status.

One hundred eighty-three children were considered for inclusion in the study. Of these, 33 had undergone nonfocal surgical procedures (19 hemispherectomy, three disconnections, three shunt, eight vagus nerve stimulator) and were excluded. Six additional patients were excluded due to inability to administer Wechsler IQ assessments at baseline or follow-up (three at age < 5 years at presurgical assessment, two nonfluent in English, one deaf).

Of the 144 remaining children meeting above criteria, two were deceased at follow-up, and 43 were noncontactable. Of the remaining 99 individuals who were contacted, 32 chose not to participate and 14 were unable to take part (pregnant, $n = 2$; disability, $n = 10$; had moved overseas, $n = 2$). One participant dropped out before completing IQ assessments at follow-up. Fifty-two participants were recruited: 39 who underwent focal resection and 13 treated with medication only. Additional detail on participant characteristics are provided in supplementary Table S1.

The London Queen Square Ethics Committee approved this study. Consent was obtained from participants aged ≥ 16 years at the time of follow-up evaluation. Assent from participants and consent from parents was obtained from those younger than 16 years.

2.2 | Procedures

Presurgical assessments included MRI, ictal and interictal electroencephalogram, and neuropsychological assessment. Age-appropriate IQ assessments were completed using the Wechsler intelligence scales, including Wechsler Preschool and Primary Scale of Intelligence (WPPSI-II, WPPSI-R), Wechsler Intelligence Scales for Children (WISC-III, WISC-IV), Wechsler Adult Intelligence Scales (WAIS-III, WAIS-IV), and Wechsler Abbreviated Scale of Intelligence. Baseline refers to the assessment carried out prior to surgery. In those who underwent repeat surgery ($n = 4$), baseline

preceded the first surgery. For nonsurgical patients, baseline refers to their first assessment at GOSH.

Baseline FSIQ data were available in all patients. Component scaled scores included verbal comprehension index (VCI; available in $n = 48$), processing speed index (PSI; $n = 44$), related indices for perceptual reasoning and perceptual organization (PRI/POI; $n = 48$), and related indices for working memory and freedom from distractibility (WMI/FFDI; $n = 45$). Incomplete baseline data were due to preschool and abbreviated tests not providing subscale data ($n = 4$) or insufficient subtests completed ($n = 4$). Supplementary pre- or postoperative IQ data were available in 70% ($n = 38$), most frequently at 1-year follow-up (total number of IQ assessments available = 161).

Follow-up assessment was completed from June 2012 to April 2016, including IQ assessment and structural MRI. Surgical cases and controls were recruited concurrently. Follow-up was on average 7.2 years after surgery (range = 1.6–15.7) and 7.7 years after baseline assessment (range = 2.5–16). Age-appropriate IQ tests were carried out (WISC-IV for age ≤ 16 years and 2 months, WAIS-IV for all others).

Volumetric preoperative MRI scans were available for 37 focal surgery and 12 nonsurgical patients (94.2% of total sample). MRI scans were obtained from all participants at follow-up. All MRI scans acquired prior to 2007 ($n = 20$, 41% of scans at baseline, primarily in the surgical sample [$n = 19$]) were obtained on a 1.5-T Siemens Vision System and included a volume T1-weighted scan using a three-dimensional magnetization-prepared rapid gradient echo sequence (repetition time = 10 milliseconds, echo time = 4 milliseconds, flip angle = 12° , voxel size = $1.0 \times 1.0 \times 1.25$ mm). Scans acquired after 2007, including all follow-up scans, were obtained on a 1.5-T Siemens Avanto System. Three-dimensional volume T1-weighted scans were acquired using a 3DFLASH sequence (repetition time = 11 milliseconds, echo time = 5 milliseconds, flip angle = 15° , voxel size = $1.0 \times 1.0 \times 1.0$ mm).

2.3 | MRI analysis

Radiological diagnoses were given by experienced pediatric neuroradiologists, who evaluated all preoperative MRI images. Locations of resections or lesions were categorized by the research team as follows: frontal-central, parietal, occipital, temporal, or multilobar abnormalities/resections. Resections were manually traced in native space on postoperative T1-weighted MRI images, to provide resection maps and derive resection volumes. Tracings were completed with reference to the contralesional hemisphere using MRICron (<https://people.cas.sc.edu/rorden/mricron/index.html>) by one researcher (C.S.).

Resection maps were normalized to the International Consortium for Brain Mapping (ICBM) space template

in SPM12 (<http://www.fil.ion.ucl.ac.uk/spm/software/spm12/>) and entered into voxel-based lesion symptom mapping analysis (VLSM)²¹ using npm (<https://www.mccauslandcenter.sc.edu/crnl/tools>).²² Voxelwise *t* tests evaluated resections associated with change in IQ (postoperative – preoperative IQ score) in regions of the brain resected in ≥ 2 individuals. The level of significance was set at $\leq .05$ with false discovery rate (FDR) correction for multiple comparisons.

Images were segmented into white matter (WM), gray matter (GM), and cerebrospinal fluid (CSF) using the Computational Anatomy Toolbox 12 (<http://dbm.neuro.uni-jena.de/cat/>) for SPM12 and normalized into ICBM space. This segmentation procedure estimates tissue probabilities without priors and uses a partial volume segmentation approach to estimate the amount of each tissue type in each voxel. Quality estimates are provided for each scan, and one baseline scan ($< 65\%$ quality) was excluded from analyses. GM and WM volumes were extracted for ipsilesional and contralesional hemispheres. Summation of GM, WM, and CSF volumes provided total intracranial volume (TIV).

Maps of brain volume change were obtained by subtracting normalized preoperative GM and WM segmented images from those obtained at follow-up using the *imcalc* function in SPM12. Images were smoothed using a 10-mm full-width at half-maximum Gaussian kernel. Voxelwise regression examined FSIQ change in relation to contralesional GM and WM volume changes, covarying for TIV change, sex, and age at follow-up. A voxelwise threshold of $P < 0.05$ with familywise error (FWE) correction was used for the entire contralesional hemisphere for right and left surgery groups separately. Regions of interest implicated in IQ included the following contralesional brain structures: dorsolateral prefrontal cortex (Brodmann areas [BAs] 6, 9, 10, 45, 46, 47), inferior (BAs 39, 40) and superior (BA 7) parietal lobule, the anterior cingulate (BA 32), and temporal (BAs 21, 37) and occipital regions (BAs 18, 19).²³ These regions were evaluated at a $P < 0.001$ uncorrected threshold with cluster size of ≥ 200 .

2.4 | Statistical analysis

Change indices were calculated by subtracting baseline from follow-up values (positive values index increases, negative values index decrements). Clinical and brain structural factors associated with postsurgical outcome were examined in the surgical cohort only. Analyses were completed in SPSS version 24 and SAS 9.4.

All data were assessed for normality using the Shapiro-Wilk statistic, followed by parametric and nonparametric tests, as appropriate. Frequencies are reported with 95%

confidence intervals (CIs), derived from bootstrapping with simple sampling in 1000 samples, presented in square brackets.

A threshold of gains or losses of 10 FSIQ points was applied to examine individual change, in line with previous research^{8,13,24,25} and in keeping with broadly applied ranges for categorizing cognitive ability.²⁶

Group-level analysis examined differences in epilepsy-related and clinical characteristics (seizure onset, duration of epilepsy, seizure frequency, medication cessation, baseline IQ, duration of follow-up) between surgical and nonsurgical groups. Furthermore, in the surgical group the effect of clinical characteristics was examined on FSIQ at baseline, at follow-up, and change over time. Analyses were completed with *t* tests, Mann-Whitney *U* tests, analysis of variance (ANOVA), Kruskal-Wallis, Wilcoxon signed-rank, chi-square, and Fisher's exact tests, and bivariate and partial correlations, as required.

Longitudinal analyses comparing pre- to postoperative FSIQ change between surgical and nonsurgical groups were conducted with repeated-measures ANOVA. These analyses were repeated as an analysis of covariance after including as covariates demographic measures that differed or showed a trending difference between groups at baseline. Change in individual component scaled scores over time were examined with a $2 \times 2 \times 4$ repeated measures ANOVA, with independent predictors including group (surgical, nonsurgical), time (baseline, follow-up), and component scaled scores (VCI, PRI/POI, WMI/FFDI, PSI). Departures from sphericity were examined, and Greenhouse-Geisser corrections were adopted where required.

To examine longitudinal change over all available FSIQ data, including supplementary pre- and postoperative assessments, a linear mixed model with random intercept and restricted (residual) maximum likelihood estimation procedure was carried out with the SAS software package. This allowed test results to account for correlated observations nested within individuals. FSIQ was entered as dependent variable, and independent variables included surgical status at time of assessment (did not [yet] undergo surgical treatment vs treated surgically), duration of time from baseline (time), and time $\times$ surgical status interaction. Test type (child vs adult test) was included as an independent predictor, because transition from child to adult assessments can impact on IQ scores.¹⁷

To identify the most robust predictors of FSIQ change after surgery, all clinical and brain imaging measures associated with FSIQ change, including neuroimaging predictors, were included in a multiple linear regression using the Enter method. Predictors were entered in order of their strength of association with FSIQ change.

Additional detail on participant characteristics are provided in Table S1.

3 | RESULTS

3.1 | Participants

Surgeries were completed at GOSH in 37 individuals (at age ≤ 18 years), and at NHNN in two participants (age = 18) who were recruited into the study (average age at first surgery = 14 years, range = 6-18). Comparison with a subgroup of participants in whom recruitment was unsuccessful ($n = 59$ where presurgical IQ assessments were documented in surgical databases) indicated a significant difference between groups (nonrecruited baseline IQ = 78.54, recruited baseline IQ = 86.93, $t = 2.28$, $P = 0.02$, Hedges $g = 0.44$). There were no differences in age at first assessment ($P = 0.437$), sex ($P = 0.475$), side of pathology ($P = 0.181$), lobar location of pathology ($P = 0.620$), and presence of daily seizures ($P = 0.395$) between recruitment groups.

In the nonsurgical epilepsy group, two participants were offered surgical treatment but declined. Eleven participants were not offered surgical treatment due to lack of visible lesion on MRI ($n = 2$), lesion proximity to motor cortex ($n = 2$), seizure control gained with AEDs ($n = 4$), or predicted benefits of surgery outweighed by likely deficits ($n = 3$).

Demographic and clinical data are presented in Table 1, alongside test statistics for group differences. Surgical and nonsurgical epilepsy groups did not differ with respect to sex, baseline FSIQ, or duration of epilepsy to baseline assessments. However, surgical participants had higher baseline seizure frequency and were significantly older than nonsurgical participants at both baseline and follow-up. Duration of follow-up from baseline was also longer for surgical patients. Participants were predominantly right-handed. Where available, presurgical functional MRI examination of language lateralization using a covert verb generation task and region of interest analysis centered around Broca's area (methods in Croft et al²⁷) indicated predominantly left-lateralized language (Table 1) in both surgical and nonsurgical groups.

3.2 | Seizure outcome and medication

Surgical participants were more often seizure-free at follow-up (72%; [95% CI = 63%-85%]) compared with nonsurgical participants (15% [0%-39%], Fisher's exact test, $P < 0.001$). More than half (55%) of those experiencing continuing seizures in surgical and nonsurgical groups had reduced seizure frequency. Seizure freedom was highest after frontal-central resections (100%), followed by temporal (77%), multilobar (50%), and parietal (43%). Two of four participants who underwent repeat surgery for continuing seizures were seizure-free at follow-up. Fifty percent of surgical patients were no longer taking AEDs at follow [35%-69%], compared with 8% in the nonsurgical group ([0%-23%]; Fisher's exact test: $P = 0.008$).

3.3 | FSIQ and component scaled scores

FSIQ gains of at least 10 points were observed in 15 (39% [23%-54%]) surgical patients and one (8% [0%-23%]) nonsurgical patient (range = 10-28 point change). Losses (range -10 to -27) were seen in four surgical patients (10% [3%-21%]) and two nonsurgical patients (15% [0%-39%]). One participant with decline in FSIQ was psychotic at follow-up and excluded from further analysis of neuropsychological assessments.

Groups showed divergent trajectories between baseline and follow-up, with surgical patients showing modest increases in FSIQ, whereas nonsurgical controls showed no overall change (Figure 1A; group by time interaction ($F_{1, 49} = 5.93$, $P = 0.024$). Repeating analyses after covarying for group differences at baseline (age at baseline assessment, baseline seizure frequency, and age at first seizure) resulted in a reduced significance level, which failed to meet statistical thresholds ($F_{1, 46} = 3.28$, $P = 0.08$).

Linear mixed modeling of FSIQ over all available assessments showed a reduction of IQ over time (estimate = -1.6, SE = 0.51, $t = -3.11$, $P = 0.002$) and no main effect of surgical status (estimate = -0.61, SE = 1.83, $t = -0.33$, $P = 0.74$). However, a significant time by surgical status interaction effect (estimate = 0.94, SE = 0.31, $t = 3.04$, $P = 0.003$) was seen, reflecting a changing trajectory of FSIQ over time from before to after surgery. No main effects of child versus adult test type were seen (estimate = -2.50, SE = 1.97, $t = -1.27$, $P = 0.21$). Findings in the surgery group were visualized using a locally weighted scatter plot smoothing regression curve of residuals (regressing out baseline FSIQ and test type, $\alpha = .05$, smoothing parameter = 80%). Figure 1C shows results, where time = 0 is centered at time of surgery.

Change in IQ component scaled scores (VCI, PRI/POI, WMI/FFDI, and PSI) from baseline to follow-up were examined with repeated measures ANOVA. This showed a significant interaction between group and time of assessment (Greenhouse-Geisser corrected $F_{2,6, 41} = 6.15$, $P = 0.017$), but no interaction between group, time of assessment, and component scales ($F_{2,6, 41} = 1.36$, $P = 0.26$). Findings reflect the modest overall improvement in IQ scaled scores over time in surgical participants (from mean of 90 to 94.8) in the context of a modest decrease in nonsurgical controls (from a mean of 92.3 to 89.4). Changes in FSIQ and component scaled scores were not correlated with the number of IQ assessments completed (range = 2-6; ρ range = 0.06-0.25, $P \geq 0.10$).

3.4 | Clinical factors associated with IQ change after surgery

FSIQ change did not differ with respect to the lobar location of resection (Figure 1B; $F_{3, 37} = 2.15$, $P = 0.11$), the side of resection ($t = -1.80$, $P = 0.08$), or the underlying pathology

TABLE 1 Clinical characteristics of epilepsy participant groups

	Participant group		Statistics, where applicable		
	Focal resection	Nonsurgical	Test statistic, z , t , χ^2 , or Fisher's exact	P	Effect size, Hedges g or Cramer V
Demographic variables					
Total number of participants	39	13			
Sex: male/female, n	21/18	7/6	0.004	0.99	0.008
Age at epilepsy onset, y , mean (SD)	6.3 (4.3)	3.9 (3.7)	3.51	0.06	0.58
Baseline characteristics					
Age at baseline, y , mean (SD)	12.9 (3.2)	10.7 (3.3)	4.16	0.04	0.68
Duration from epilepsy onset to baseline assessment, y , mean (SD)	6.5 (4.1)	6.8 (3.5)	0.22	0.83	0.06
Baseline FSIQ, mean (SD)	86.6 (18.1)	87.8 (14.3)	0.23	0.83	0.07
Baseline seizure frequency, n , daily/weekly/monthly/<monthly	24/10/3/2	1/3/7/1 ^a	—	0.001 ^b	—
Handedness, n , right/left/ambidextrous	28/10/1	7/5/1	1.69	0.45	0.43
Language lateralization, n , typical/atypical ^c	24/4	9/2			
Follow-up characteristics					
Age at follow-up, y , mean (SD)	21.2 (4.9)	16.6 (3.9)	3.07	0.003	0.98
Duration between baseline and follow-up assessment, y , mean (SD)	8.3 (3.8)	5.9 (2.1)	3.99	0.05	0.72
Pathology defined radiologically, n (% of group)					
MTS	3 (7.7%)	1 (7.7%)			
Dysembryoplastic neuroepithelial tumor	14 (35.9%)	3 (23.1%)			
Focal cortical dysplasia	11 (28.2%)	1 (7.7%)			
Cavernoma	4 (10.3%)	—			
Polymicrogyria	2 (5.1%)	—			
Ganglioglioma	1 (2.6%)	—			
Dual pathology, MTS+lesion	2 (5.1%)	—			
Atrophy	1 (2.6%)	2 (15.4%)			
Vascular pathology: infarct, ischemia	—	3 (23.1%)			
Sturge-Weber syndrome	—	1 (7.7%)			
Unclear pathology	1 (2.6%)	—			
No visible lesion on MRI	—	2 (15.4%)			
Resected/affected lobe, n (% of group)					
Frontal/central	9 (23.1%)	1 (7.7%)			
Parietal	7 (17.9%)	2 (15.4%)			
Temporal	17 (43.6%)	2 (15.4%)			
Multilobar	6 (15.4%)	4 (30.8%)			
Hemisphere	—	2 (15.4%)			
Resected/affected side, n , L/R/nonlocalizing	23/16/—	9/2/2		0.29 ^d	

Note. Significance was determined by chi-square test, independent samples t test, and Mann-Whitney U tests, as appropriate. Effect sizes: Hedges g is presented for continuous data and Cramer V for ordinal/dichotomous data.

Abbreviations: FSIQ, full-scale intelligence quotient; L, left; MRI, magnetic resonance imaging; MTS, mesial temporal sclerosis; R, right.

^aBaseline seizure frequency not included for one patient due to fluctuating seizure frequency.

^bFisher's exact statistic comparing frequency of daily seizures in surgical and nonsurgical groups.

^cLanguage lateralization information is provided, where available presurgically: typical = left, atypical = right or bilateral.

^dFisher's exact statistic comparing frequency of left and right affected side in surgical and nonsurgical groups.

after restriction to the four main pathology types in this sample (dysembryoplastic neuroepithelial tumors, focal cortical dysplasia, mesial temporal sclerosis, and cavernoma; $F_{3,30} = 1.65$, $P = 0.20$).

In surgical participants, FSIQ change was inversely correlated with baseline FSIQ ($r = -0.34$, $P < 0.04$), with greater increases in those who had lower IQs at baseline. Neither seizure outcomes (seizure-free vs continuing seizures: $F_{1,36} = 2.38$, $P = 0.13$), nor AED discontinuation ($F_{1,35} = 3.19$, $P = 0.08$) was found to influence IQ change over time. Participants who were seizure-free and had discontinued AED treatment at follow-up had higher FSIQ at both baseline and follow-up compared to those with continuing seizures or continued AED treatment (seizure cessation: $F_{1,36} = 13.63$, $P = 0.001$; AED discontinuation: $F_{1,35} = 5.67$, $P = 0.02$).

Age at seizure onset was correlated with FSIQ at baseline ($\rho = 0.50$, $P = 0.001$) and follow-up ($\rho = 0.54$, $P = 0.001$), but not FSIQ change ($\rho = 0.025$, $P = 0.88$). Preoperative epilepsy duration (time between onset and surgery) was inversely correlated with FSIQ at baseline ($r = -0.48$, $P = 0.001$) and follow-up ($r = -0.46$, $P = 0.004$), but not FSIQ change ($r = 0.05$, $P = 0.77$). Age at surgery did not correlate significantly with FSIQ at baseline, follow-up, or FSIQ change ($\rho = -0.01$ to 0.03 , $P \geq 0.84$). Longer follow-up was associated with greater IQ improvement ($\rho = 0.38$, $P < 0.02$).

3.5 | Brain imaging measures associated with IQ change

Resection volumes varied in size ($1\text{--}66\text{ cm}^3$) and showed a trending inverse association with FSIQ change ($\rho = -0.30$, $P = 0.07$). VLSM showed that FSIQ decline was more common after resections to the left superior and middle temporal gyri ($z = 3.67$, $P < 0.05$ FDR corrected; Figure 2B). VCI reductions were associated with resections to the left temporal pole extending through to the midtemporal lobe including mesial temporal structures ($z = 2.35$, $P < 0.05$ FDR corrected; Figure 2B). Three left temporal surgery patients with unchanged or improved VCI underwent smaller, more posterior, or more lateral resections than those cases with VCI declines (Figure 2C). No voxels were associated with change in other component scaled IQ scores (WMI/FFDI, PRI/POI, or PSI).

In the surgery group, FSIQ change positively correlated with change in total GM volume ($r = 0.59$, $P < 0.001$, Figure 3A) but not WM volume ($r = -0.09$, $P = 0.61$). This correlation remained significant after covarying for transition between MRI scanners (partial correlation $r = 0.61$, $P < 0.001$) and was also present when examining separately individuals in whom scanner transition had occurred ($r = 0.53$, $P = 0.03$, $n = 18$) and those for whom the same MRI scanner

and imaging parameters had been used at both baseline and follow-up ($r = 0.70$, $P = 0.002$, $n = 17$). This correlation remained significant when examining GM volumes in both contralesional (controlling for age at follow-up and sex: $r = 0.48$, $P = 0.003$) and ipsilesional hemispheres (additionally covarying for resection volume: $r = 0.55$, $P < 0.001$).

Change in brain GM volume was not associated with seizure cessation ($t = -0.04$, $P = 0.97$) or AED discontinuation ($t = 1.26$, $P = 0.22$). Examining the variables associated with IQ change in relation to GM volume change revealed no significant correlations (baseline IQ: $r = -0.19$, $P = 0.28$; duration of follow-up: $\rho = 0.05$, $P = 0.79$).

Voxelwise regression analysis identified a peak T value of 9.1 in the right middle frontal gyrus (MFG; BA 46; coordinates 34, 50, 26; cluster size 231) in individuals with surgery to the left hemisphere. Here, a positive association between GM density change and FSIQ change was seen at the $P = 0.003$ (FWE corrected) threshold. Within predefined regions of interest, additional clusters in frontal, temporal, and occipital lobes emerged (Figure 3B, $P \leq 0.001$ uncorrected). No significant contralesional volume changes were seen after surgery to the right hemisphere, nor for investigations of contralesional WM volume.

Parameter estimates for GM change in the contralesional MFG cluster were extracted for participants with surgery to left and right hemispheres. These correlated with FSIQ change across both left and right surgical groups ($r = 0.59$, $P < 0.001$), as well as change in perceptual reasoning/perceptual organization indices (PRI/POI: $r = 0.51$, $P = 0.003$), but not with change in other component scaled scores (VCI, WMI/FFDI, or PSI: $r \leq 0.24$, $P \geq 0.20$). Associations between FSIQ and PRI/POI change and the parameter estimates were driven primarily by data from individuals with surgery in the left hemisphere. Although these associations were non-significant in the right hemispheric focal surgery group, the direction of effect was similar (FSIQ change: $r = 0.38$, $P = 0.20$; PRI/POI change: $r = 0.46$, $P = 0.12$).

3.6 | Predictors of IQ change after surgery

To identify the most robust predictors of FSIQ change after surgery, all clinical and brain imaging measures associated with FSIQ change were compared using multiple linear regression. Predictors were entered in order of their strength of association with FSIQ change: total GM volume change, location of resection (anterior left temporal or elsewhere), time between baseline and follow-up, and preoperative IQ. The regression model was significant ($F_{4,34} = 8.45$, $P < 0.001$, $R^2 = 0.53$), with two significant predictors. GM volume increases predicted a positive IQ change ($\beta = .45$, $P = 0.002$); left anterior temporal resections predicted a negative change ($\beta = -.38$, $P = 0.01$).

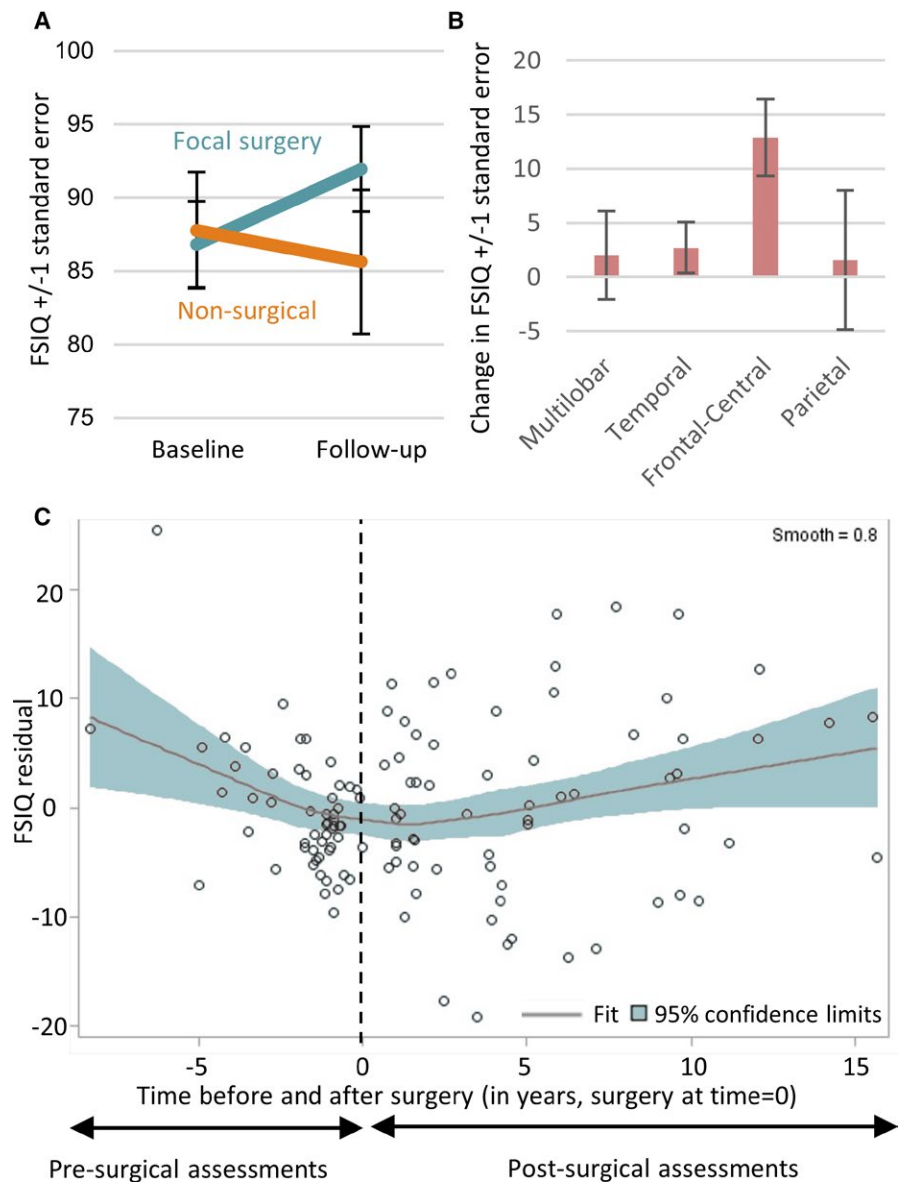


FIGURE 1 Intelligence quotient (IQ) change over time. A, Mean full-scale IQ (FSIQ) of surgical and nonsurgical groups from baseline to follow-up, showing interaction of group by time effect. B, Change in FSIQ over period of follow-up (difference postsurgery-presurgery) by resection type. C, Local regression graph with fit line, depicting all available FSIQ scores in the surgical group (residuals after regressing out preoperative FSIQ and adult/child test transition) with 95% confidence intervals. Regression line reflects gradual increase in FSIQ from time of surgery (centered at $x = 0$) over time

4 | DISCUSSION

Children who underwent focal surgery for epilepsy showed modest groupwise improvement in IQ; this outcome was not observed in a group of children with severe childhood epilepsy who did not undergo epilepsy surgery. However, postsurgical improvement occurred in the context of significant individual variation. Using voxel-based analysis methods, we showed that a combination of brain growth and surgical factors may have contributed to IQ change.

Researchers have speculated that eliminating the effects of seizures and AEDs from the developing brain may reduce maturational disturbances in functional cortical networks, playing a critical part in postsurgical improvement.⁴ The current research supports this hypothesis and extends previous findings⁹ to a much broader group, including children

with extratemporal surgery and a variety of pathologies. We find change in IQ to be associated with global GM volume changes and localized changes within the contralesional prefrontal cortex, an association region maturing later in adolescence^{28,29} and closely implicated in performance on cognitive assessments of IQ.²³

Overall, a reduction in GM volume from pre- to postsurgical assessment was seen in 83% of participants who underwent surgery. This likely reflects the resections themselves in combination with maturational decreases in cortical thickness from childhood into adulthood.³⁰ Research has shown that developmental trajectories in cortical thickness, in frontal regions in particular, vary according to intellectual development of the study subjects.³¹ Whereas local cortical thickness and volume show a primarily linear decline with age during this time period, local cortical surface area shows quadratic

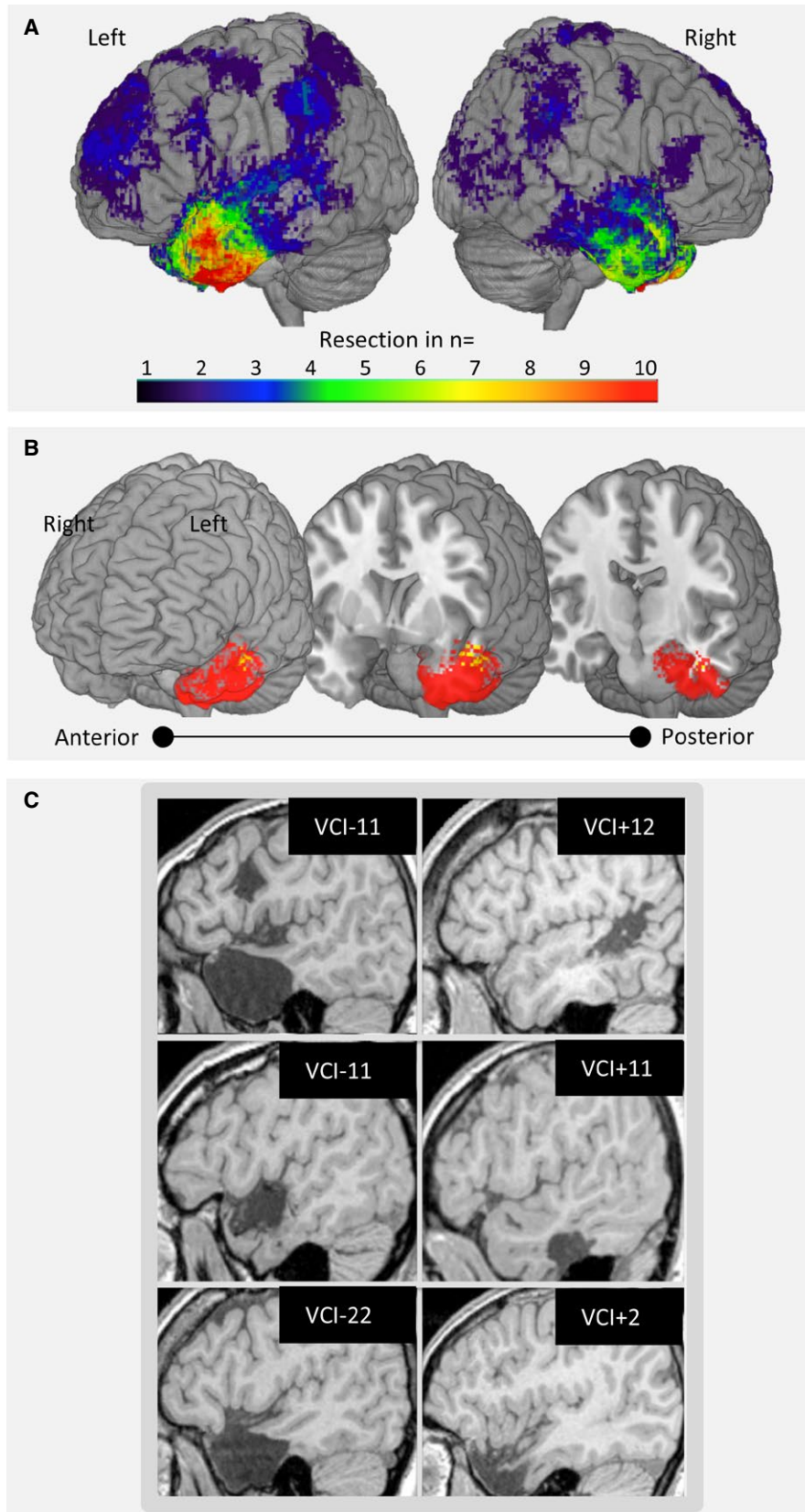


FIGURE 2 A, Resection density map showing frequency of resection locations in patients with surgery. B, Results from voxel-based lesion symptom mapping ($P < 0.05$ false discovery rate corrected). Highlighted in red are resections associated with declines in verbal comprehension index (VCI); in yellow are resections associated with declines in full-scale intelligence quotient and VCI. C, Magnetic resonance imaging scans for three patients without VCI decline, and three with the largest decline after left temporal surgery

growth in a range of brain regions including the prefrontal cortex (increasing until adolescence and then declining).³² Moreover, our findings are in agreement with research showing that global and local GM volume is correlated with IQ.^{33,34}

Verbal reasoning declined in children with surgery to the left anterior temporal lobe, a brain region implicated in semantic memory functions in adults.³⁵ Results indicate reduced reorganizational capacity for semantic functions,

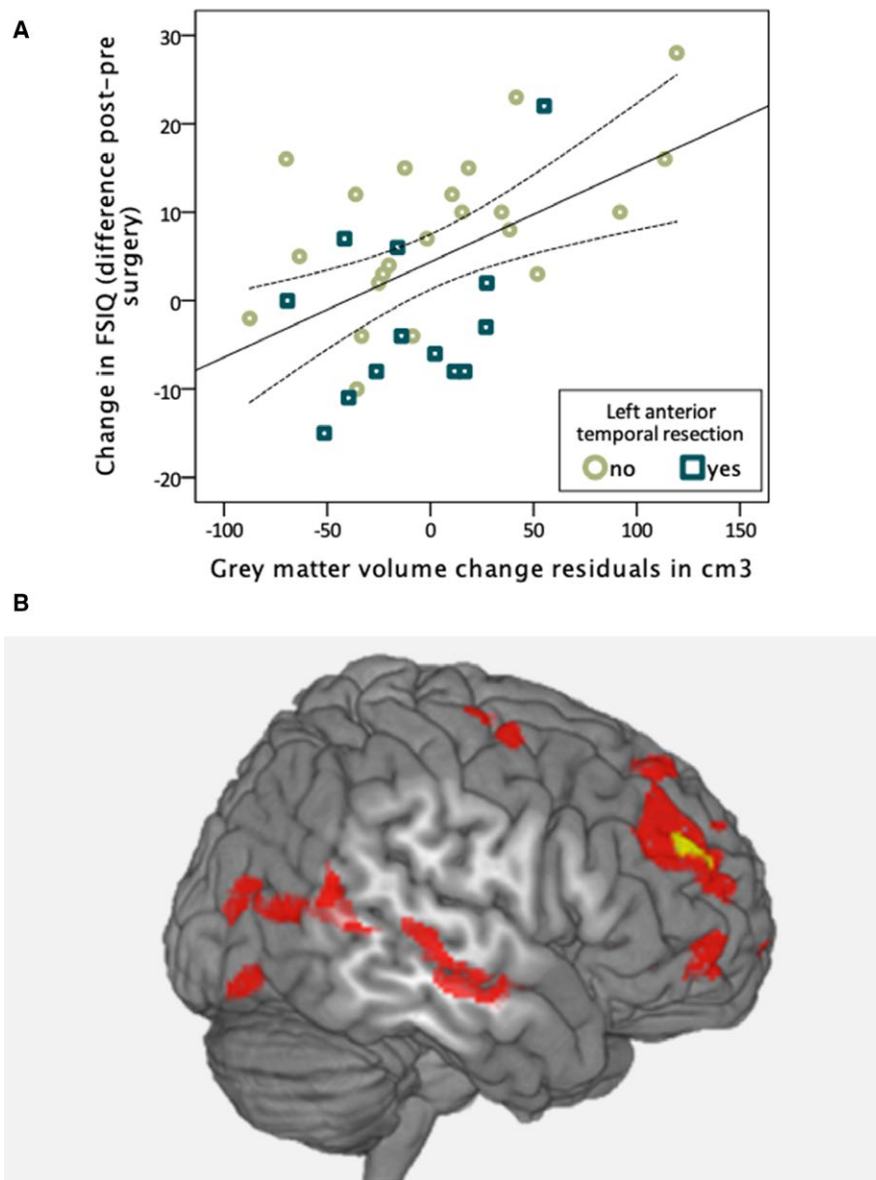


FIGURE 3 Gray matter volume change after surgery. A, Changes in total gray matter volume (residuals after regressing out age, sex, duration of epilepsy and resection volume) in relation to full-scale intelligence quotient (FSIQ) change in the surgery group, with regression line and 95% confidence intervals of the mean. B, Right hemisphere regions where increase in focal gray matter volume is positively correlated with gain in FSIQ after left hemisphere surgery (yellow, $P < 0.05$ familywise error corrected, red $P < 0.001$ uncorrected)

broadly in line with results from a previous study showing that better semantic memory is associated with greater temporal pole integrity after pediatric temporal lobe surgery.²⁰ Further work is now required to examine underpinnings of this decline, to elucidate whether specific deficits, such as naming or word retrieval problems, may be influencing these outcomes (eg, Busch et al³⁶).

The current data show that effects of cognitive improvement due to cortical growth and restrictions in verbal cognitive development with anterior temporal resections can co-occur within the same individual. As shown in Figure 3, one individual with anterior temporal resection showed an increase of 22 IQ points over time. This individual

experienced a decline in verbal IQ as associated with anterior temporal resection (VCI decrease from 77 to 70) and simultaneously an increase in performance IQ measures (increase in PRI/POI from 63 to 104) and significant cortical growth. Similar effects were seen in two other individuals with anterior temporal lobe resections (VCI = -9, PRI/POI = +19; VCI = -9, PRI/POI = +8). This indicates that changes seen are not simply attributable to a global cognitive improvement or decline.

Replicating previous results, longer follow-up was associated with greater postsurgical improvement, highlighting the importance of carrying out routine long-term follow-up assessments to identify reliable changes.^{8,9} No association was

seen between IQ change and age at surgery. However, this finding does not clarify the potential cost of delaying intervention for cognitive outcomes.³⁷ The average duration between epilepsy onset and surgery in our study was 7.5 years. Shorter duration of epilepsy has been associated with better presurgical cognitive functioning³⁸ and better long-term seizure control.³⁹ However, because age at onset, duration of epilepsy, and age at surgery are intercorrelated, larger cohorts are required to clarify interactions between these factors to identify critical periods for intervention and optimized outcomes.

In contrast to previous studies, IQ change in this cohort was not significantly influenced by postsurgical cessation of seizures^{3,8} or AED treatment.^{4,9,15} The more limited postsurgical use of topiramate in the current sample may account for differences in relation to other cohorts (eg, Skirrow et al⁹). Negative impacts of topiramate on cognition have been identified in previous randomized studies.^{40,41} Furthermore, the effects of seizure and AED reduction/cessation may be graded rather than categorical; in the current study, patients who became seizure- and AED-free ($n = 20$) showed an average nine-point increase in FSIQ, compared to +4 in seizure-free patients who continued AED treatment ($n = 7$), +2 for those with reduced seizure frequency ($n = 6$), and a six-point decrease in those with unchanged seizure frequency ($n = 5$).

Although practice effects have been noted for IQ assessments, in particular for performance IQ measures (PRI/POI),⁴² changes in FSIQ or component scaled scores were not correlated with the number of IQ assessments conducted with each individual. Practice effects may have been limited in the current study due to the typically long duration between assessments (median = 3.4 years), which is likely to have reduced carryover or learning effects.

Our participants were selected on the basis of available preoperative cognitive data and availability for neuroimaging and cognitive assessments. We noted a modest but significant difference of IQ at baseline between individuals who were recruited into the study and those who were not. It is not clear whether these differences were due to greater difficulty for some participants to attend follow-up investigations (perhaps due to greater inflexibility of work, educational or support arrangements, or more difficulty with travel and participating in cognitive and MRI assessments). Efforts are now required to promote inclusivity, for example through mobile or home-based assessment options, and more flexible or proximal options for MRI, to promote broader participation and further improve accuracy of outcome data.

Although a waiting list randomization study of pediatric epilepsy surgery with a 12-month follow-up has recently been published,⁴³ it is not feasible to examine longer-term outcomes in this manner.^{8,13} Children matched for basic

illness characteristics provide a suitable comparison group for estimating developmental trajectory in epilepsy without surgical intervention.¹⁵ Although there was in part overlapping pathological diagnoses between surgical and nonsurgical participants, participants in the nonsurgical group also had pathological diagnoses that were not represented in our surgical sample. This control group allows us to control for potential sources of bias in our results, including the impact of epilepsy-related developmental variables, and the effects of regression to the mean in repeated assessments. However, the nonsurgical group also differed from the surgical sample on a number of characteristics, including lower baseline seizure severity and younger age at follow-up. Although this study provides insights into postsurgical IQ development, replication is required, and multicenter studies are needed to clarify outcomes where a variety of interacting or additive factors are operating.

ACKNOWLEDGMENTS

We thank our participants and their families, who made this work possible. We also thank the GOSH epilepsy surgery, neurology, neuropsychology, and neuroradiology departments, and the Great Ormond Street Institute of Child Health Developmental Imaging and Biophysics Section.

CONFLICT OF INTEREST

J. Helen Cross reports grants from Vitaflo; honoraria from UCB, Nutricia, GW Pharma, and Zogenix, all paid to the department; and participation as investigator in clinical trials for Zogenix, GW Pharma, Marinus, and Takeda. Caroline Skirrow is employed by Cambridge Cognition. All other authors have nothing to disclose. We confirm that we have read the Journal's position on issues involved in ethical publication and affirm that this report is consistent with those guidelines. Data are reported following STROBE case-control guidelines, identified as the most appropriate reporting guideline through the Equator Wizard (<https://www.penelope.ai/equator-wizard>). Selected data can be obtained from the authors on request.

AUTHOR CONTRIBUTIONS

Study conception and design: Cross, Baldeweg, and Vargha-Khadem. Data collection and analysis: all authors. Writing manuscript and critical revision: all authors.

ORCID

Caroline Skirrow  <https://orcid.org/0000-0001-8692-7787>

Torsten Baldeweg  <https://orcid.org/0000-0002-5724-1679>

REFERENCES

- Wiebe S, Jette N. Pharmacoresistance and the role of surgery in difficult to treat epilepsy. *Nat Rev Neurol*. 2012;8(12):669–77.
- Téllez-Zenteno JF, Dhar R, Wiebe S. Long-term seizure outcomes following epilepsy surgery: a systematic review and meta-analysis. *Brain*. 2005;128(5):1188–98.
- Battaglia D, Chieffo D, Lettori D, Perrino F, Di Rocco C, Guzzetta F. Cognitive assessment in epilepsy surgery of children. *Childs Nerv Syst*. 2006;22(8):744–59.
- Hallböök T, Tideman P, Rosén I, Lundgren J, Tideman E. Epilepsy surgery in children with drug-resistant epilepsy, a long-term follow-up. *Acta Neurol Scand*. 2013;128(6):414–21.
- Battaglia D, Chieffo D, Tamburrini G, Lettori D, Losito E, Leo G, et al. Posterior resection for childhood lesional epilepsy: neuropsychological evolution. *Epilepsy Behav*. 2012;23(2):131–7.
- Viggedal G, Kristjansdóttir R, Olsson I, Rydenhag B, Uvebrant P. Cognitive development from two to ten years after pediatric epilepsy surgery. *Epilepsy Behav*. 2012;25(1):2–8.
- Ramantani G, Kadish NE, Mayer H, et al. Frontal lobe epilepsy surgery in childhood and adolescence: predictors of long-term seizure freedom, overall cognitive and adaptive functioning. *Clin Neurosurg*. 2018;83(1):93–102.
- Puka K, Tavares TP, Smith ML. Development of intelligence 4 to 11 years after paediatric epilepsy surgery. *J Neuropsychol*. 2017;11:161–73.
- Skirrow C, Cross JH, Cormack F, Harkness W, Vargha-Khadem F, Baldeweg T. Long-term intellectual outcome after temporal lobe surgery in childhood. *Neurology*. 2011;76:1330–7.
- Lewis DV, Thompson RJJ, Santos CC, et al. Outcome of temporal lobectomy in adolescents. *J Epilepsy*. 1996;9:198–205.
- Sibilia V, Barba C, Metitieri T, et al. Cognitive outcome after epilepsy surgery in children: a controlled longitudinal study. *Epilepsy Behav*. 2017;73:23–30.
- Ehrstedt C, Rydell AM, Gabert Hallsten M, Strömberg B, Ahlsten G. Cognition, health-related quality of life, and mood in children and young adults diagnosed with a glioneuronal tumor in childhood. *Epilepsy Behav*. 2018;83:59–66.
- Flint AE, Waterman M, Bowmer G, Vadlamani G, Chumas P, Morrall MCHJ. Neuropsychological outcomes following paediatric temporal lobe surgery for epilepsies: evidence from a systematic review. *Seizure*. 2017;52:89–116.
- Baldeweg T, Skirrow C. Long-term cognitive outcomes after epilepsy surgery in children. In: *Long-Term Outcomes of Epilepsy Surgery in Adults and Children*. Mälgren K, Cross H, Baxendale S, eds. Cham, Switzerland: Springer; 2015:85–101.
- Boshuisen K, van Schooneveld MM, Uiterwaal CS, et al. Intelligence quotient improves after antiepileptic drug withdrawal following pediatric epilepsy surgery. *Ann Neurol*. 2015;78(1):104–14.
- Roulet-Perez E, Davidoff V, Mayor-Dubois C, et al. Impact of severe epilepsy on development: recovery potential after successful early epilepsy surgery. *Epilepsia*. 2010;51(7):1266–76.
- Hermann BP, Seidenberg M, Bell B. The neurodevelopmental impact of childhood onset temporal lobe epilepsy on brain structure and function and the risk of progressive cognitive effects. *Prog Brain Res*. 2002;135:429–38.
- Blanc F, Martinian L, Liagkouras I, Catarino C, Sisodiya SM, Thom M. Investigation of widespread neocortical pathology associated with hippocampal sclerosis in epilepsy: a postmortem study. *Epilepsia*. 2011;52(1):10–21.
- Tosun D, Dabbs K, Caplan R, et al. Deformation-based morphometry of prospective neurodevelopmental changes in new onset paediatric epilepsy. *Brain*. 2011;134(4):1003–14.
- Skirrow C, Cross JH, Harrison S, et al. Temporal lobe surgery in childhood and neuroanatomical predictors of long-term declarative memory outcome. *Brain*. 2015;138(1):80–93.
- Bates E, Wilson SM, Saygin AP. Lesion–symptom mapping. *Nat Neurosci*. 2003;6(5):448–50.
- Rorden C, Karnath H-O, Bonilha L. Improving lesion-symptom mapping. *J Cogn Neurosci*. 2007;19(7):1081–8.
- Jung RE, Haier RJ. The Parieto-Frontal Integration Theory (P-FIT) of intelligence: converging neuroimaging evidence. *Behav Brain Sci*. 2007;30(2):135–54.
- Freitag H, Tuxhorn I. Cognitive function in preschool children after epilepsy surgery: rationale for early intervention. *Epilepsia*. 2005;46(4):561–7.
- Van Schooneveld MMJ, Braun KPI. Cognitive outcome after epilepsy surgery in children. *Brain Dev*. 2013;35(8):721–9.
- Dombrowski SC. *Psychoeducational Assessment and Report Writing*. New York, NY: Springer; 2015.
- Croft LJ, Rankin PM, Liégeois F, et al. To speak, or not to speak? The feasibility of imaging overt speech in children with epilepsy. *Epilepsy Res*. 2013;107(1–2):195–9.
- Sowell ER, Peterson BS, Thompson PM, Welcome SE, Henkenius AL, Toga AW. Mapping cortical change across the human life span. *Nat Neurosci*. 2003;6(3):309–15.
- Casey BJ, Tottenham N, Liston C, Durston S. Imaging the developing brain: what have we learned about cognitive development? *Trends Cogn Sci*. 2005;9(3):104–10.
- Gogtay N, Giedd JN, Lusk L, et al. Dynamic mapping of human cortical development during childhood through early adulthood. *Proc Natl Acad Sci U S A*. 2004;101(21):8174–9.
- Shaw P, Greenstein D, Lerch J, et al. Intellectual ability and cortical development in children and adolescents. *Nature*. 2006;440:676–9.
- Ducharme S, Albaugh MD, Nguyen TV, et al. Trajectories of cortical surface area and cortical volume maturation in normal brain development. *Data Brief*. 2015;5:929–38.
- Frangou S, Chitins X, Williams SCR. Mapping IQ and gray matter density in healthy young people. *Neuroimage*. 2004;23(3):800–5.
- Wilke M, Sohn JH, Byars AW, Holland SK. Bright spots: correlations of gray matter volume with IQ in a normal pediatric population. *Neuroimage*. 2003;20(1):202–15.
- Patterson K, Nestor PJ, Rogers TT. Where do you know what you know? The representation of semantic knowledge in the human brain. *Nat Rev Neurosci*. 2007;8:976–88.
- Busch RM, Chapin JS, Haut JS, Dulay MF, Naugle RI, Najm I. Word-finding difficulties confound performance on verbal cognitive measures in adults with intractable left temporal lobe epilepsy. *Epilepsia*. 2013;54(3):2011–4.
- Berg A. Paediatric epilepsy surgery: making the best of a tough situation. *Brain*. 2015;138:4–5.
- Ramantani G, Kadish NE, Anastasopoulos C, et al. Epilepsy surgery for glioneuronal tumors in childhood: avoid loss of time. *Neurosurgery*. 2014;74(6):648–57.
- Simasathien T, Vadera S, Najm I, Gupta A, Bingaman W, Jehi L. Improved outcomes with earlier surgery for intractable frontal lobe epilepsy. *Ann Neurol*. 2013;73:646–54.

40. Blum D, Meador K, Biton V, et al. Cognitive effects of lamotrigine compared with topiramate in patients with epilepsy. *Neurology*. 2006;67:400–6.
41. Loring DW, Williamson DJ, Meador KJ. Topiramate dose effects on cognition: a randomized double-blind study. *Neurology*. 2011;76:131–8.
42. Basso MR, Carona FD, Lowery N, Axelrod BN. Practice effects on the WAIS-III across 3- and 6-month intervals. *Clin Neuropsychol*. 2002;16(1):57–63.
43. Dwivedi R, Ramanujam B, Chandra PS, et al. Surgery for drug-resistant epilepsy in children. *N Engl J Med*. 2017;377(17):1639–47.

SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section at the end of the article.

How to cite this article: Skirrow C, Cross JH, Owens R, et al. Determinants of IQ outcome after focal epilepsy surgery in childhood: A longitudinal case-control neuroimaging study. *Epilepsia*. 2019;60:872–884. <https://doi.org/10.1111/epi.14707>